

$$R_I^P = [\hat{p}, \hat{q}, \hat{\omega}] \rightarrow \text{from } P \text{ to } I$$

$$R_P^I = \begin{bmatrix} \hat{p}^T \\ \hat{q}^T \\ \hat{\omega}^T \end{bmatrix} \rightarrow \text{from } I \text{ to } P$$

We can also build perifocal frame from three of the classical orbital elements: i, Ω, ω

This involves a 3-1-3 Euler angle sequence

$$R_P^I = R_3(\omega) R_1(i) R_3(\Omega)$$

$$\text{or } R_I^P = (R_P^I)^T = R_3^T(\Omega) R_1^T(i) R_3^T(\omega) = R_3(-\Omega) R_1(-i) R_3(-\omega)$$

In the perifocal frame, the position is:

$$\vec{r} = r \cos \theta \hat{p} + r \sin \theta \hat{q}$$

such that the velocity is

$$\vec{v} = [\dot{r} \cos \theta - r \dot{\theta} \sin \theta] \hat{p} + [\dot{r} \sin \theta + r \dot{\theta} \cos \theta] \hat{q}$$

Look at r

$$r = \frac{p}{1 + e \cos \theta} = p(1 + e \cos \theta)^{-1}$$

$$\dot{r} = -p(1 + e \cos \theta)^{-2} (-e \sin \theta)(\dot{\theta})$$

$$\dot{r} = \frac{p}{(1+e\cos\theta)} \cdot \frac{1}{r} \cdot e\sin\theta \left(\frac{h}{r^2}\right) \quad \text{Aside to get } \dot{\theta}$$

$$\dot{r} = r \left(\frac{h}{p}\right) e\sin\theta \left(\frac{h}{r^2}\right)$$

$$h = rv\cos\theta$$

$$h = rv_{\perp} = r(r\dot{\theta})$$

$$\dot{r} = \frac{eh}{p} \sin\theta \quad \text{recall, } p = \frac{h^2}{\mu}$$

$$\dot{\theta} = \frac{h}{r^2}$$

$$\dot{r} = \frac{e\mu}{h} \sin\theta$$

$$\text{look at } r\dot{\theta} \rightarrow r\dot{\theta} = \frac{h}{r} \quad \text{and} \quad r = \frac{\frac{h^2}{\mu}}{1+e\cos\theta} \rightarrow \frac{1}{r} = \frac{\mu}{h^2}(1+e\cos\theta)$$

$$\text{Thus } r\dot{\theta} = \frac{\mu}{h}(1+e\cos\theta)$$

Substitute back in...

$$\begin{aligned} \hat{p} &= \dot{r}\cos\theta - r\dot{\theta}\sin\theta = \left(\frac{e\mu}{h}\sin\theta\right)(\cos\theta) - \left(\frac{\mu}{h}\right)(1+e\cos\theta)(\sin\theta) \\ &= -\frac{\mu}{h}\sin\theta \hat{p} \end{aligned}$$

$$\begin{aligned} \hat{q} &= \dot{r}\sin\theta + r\dot{\theta}\cos\theta = \frac{e\mu}{h}\sin^2\theta + \left(\frac{\mu}{h}\right)(1+e\cos\theta)\cos\theta \\ &= \frac{e\mu}{h}(\sin^2\theta + \cos^2\theta) + \frac{\mu}{h}\cos\theta \\ &= \frac{\mu}{h}(e + \cos\theta) \end{aligned}$$

$$\text{Thus, } \boxed{\vec{v} = \dot{\vec{r}} = \left(-\frac{\mu}{h}\sin\theta\right)\hat{p} + \left(\frac{\mu}{h}(e + \cos\theta)\right)\hat{q}} \quad \text{equation of a circle}$$

The orbital hodograph

The hodograph is the locus of points (the curve) traced out by the tip of the vector while keeping the tail fixed at the origin.

The orbital hodograph is always a perfect circle for all types of two-body orbits: Circular, elliptical, hyperbolic, etc.

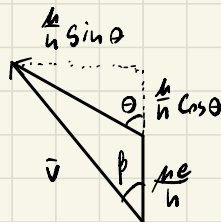
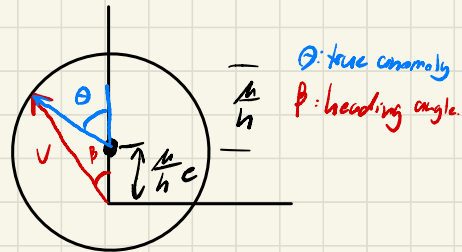
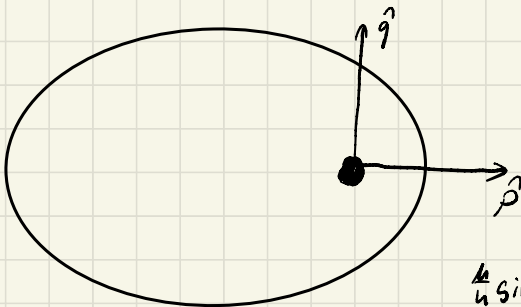
To see this more clearly

$$\vec{v} = \left(-\frac{\mu}{h} \sin \theta\right) \hat{p} + \left[\frac{\mu}{h}(e + \cos \theta)\right] \hat{q}$$

$$\vec{v}_p = \begin{bmatrix} 0 \\ \frac{\mu}{h} e \\ 0 \end{bmatrix} + \frac{\mu}{h} \begin{bmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{bmatrix}$$

↑
constant offset

↑
circle of radius $\frac{\mu}{h}$



$$R = \frac{\mu}{h}, \quad \vec{c} = \frac{\mu}{h} e \hat{q}$$

$$\tan(\beta) = \frac{\frac{\mu}{h} \sin \theta}{\frac{\mu}{h}(e + \cos \theta)}$$

$$\boxed{\beta = \frac{\sin \theta}{e + \cos \theta}}$$

We can always go from the orbital hodograph to the orbit

$$R \text{ and } \bar{C}_h \longrightarrow \Theta, \gamma, \dots \text{ etc}$$

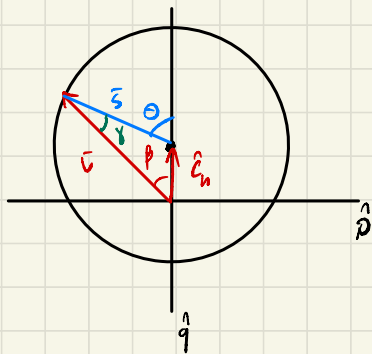
$$\text{Recall } \bar{e} = e\hat{p} \text{ and } \hat{q} \times \hat{\omega} = \hat{p}$$

$$\frac{\bar{C}_h}{R} = \frac{\frac{\mu}{h} e \hat{q}}{\frac{\mu}{h}} = e \hat{q}$$

$$\bar{e} = e\hat{p} = e\hat{q} \times \hat{\omega}$$

$$e = \frac{1}{R} \bar{C}_h \times \hat{\omega}$$

Now, consider the geometry of the hodograph



Defining $\bar{S}$ to be the vector from the hodograph center to the tip of the velocity vector

$$\bar{V} = \bar{C}_h + \bar{S}$$

$$\bar{S} = \bar{V} - \bar{C}_h$$

What is the projection of $\bar{S}$ onto $\bar{r}$?

$$\bar{r} \cdot \bar{S} = \bar{r} \cdot (\bar{V} - \bar{C}_h) = \bar{r} \cdot \bar{r} - \bar{r} \cdot \bar{C}_h$$

$\uparrow$ $\uparrow$
 r $r\dot{r}$

$$\begin{aligned} \bar{r} \cdot \bar{S} &= r\dot{r} - \bar{r} \cdot \left(\frac{\mu}{h} e \hat{q} \right) = r\dot{r} - \frac{\mu}{h} e (\bar{r} \cdot \hat{q}) & \bar{r} \cdot \hat{q} &= r \sin \theta \\ &= r\dot{r} - \frac{\mu}{h} e (r \sin \theta) \end{aligned}$$

$$\vec{r} \cdot \vec{s} = r\dot{r} - \frac{\mu}{h} e r \sin \theta$$

$$= r \left(\dot{r} - \underbrace{\frac{\mu}{h} e \sin \theta}_{\dot{r}} \right)$$

Thus, $\vec{r} \cdot \vec{s} = 0$ (orthogonal)

So, $\vec{s}$ is the local horizontal

So, the angle between $\vec{v}$ and $\vec{s}$ is γ (flight path angle)

Recall again

$$R = \frac{\mu}{h} \quad \vec{c}_n = \frac{\mu}{h} e \hat{q} = R e \hat{q}$$

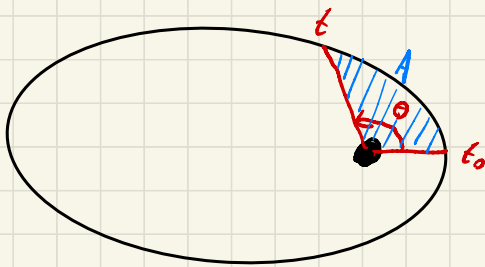
We can tell the type of orbit by looking at the hodograph.

Spacecraft Position as a function of time

Kepler's eq = Relates time and angular displacement around on orbit.

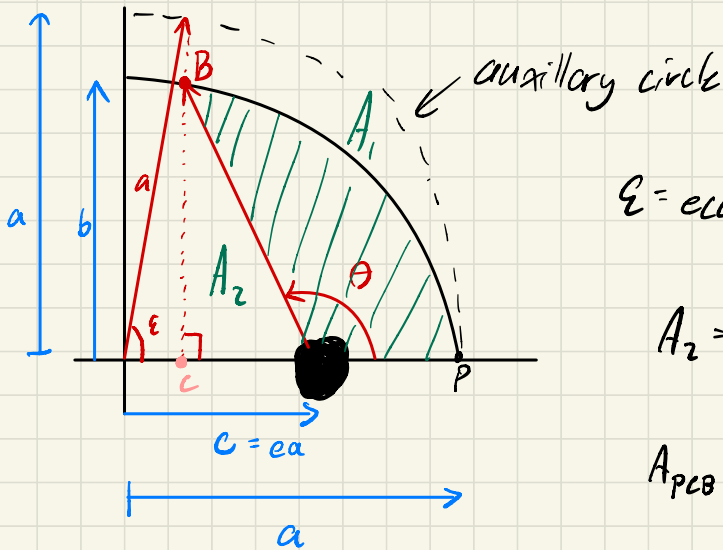
Kepler's problem: Task of finding the location of a body in orbit after a certain amount of time

We can develop these ideas geometrically: Begin with Kepler's 2nd Law.



$$\frac{t_0 - t}{P} = \frac{A}{\pi ab}$$

$$\frac{t_0 - t}{A} = \frac{P}{\pi ab}$$



ϵ = eccentric anomaly

$$A_2 = \frac{ab}{2} (\epsilon - \cos(\epsilon)) \sin(\epsilon)$$

$$A_{PCB} = \frac{ab}{2} (\epsilon - \cos(\epsilon) \sin(\epsilon))$$

$$\begin{aligned} A_1 &= A_{PCB} - A_2 \\ &= \frac{ab}{2} (\epsilon - e \sin(\epsilon)) \end{aligned}$$

Plug back in $\frac{t - t_0}{A} = \frac{P}{\pi ab}$

$$\frac{2(t - t_0)}{ab(\epsilon - e \sin(\epsilon))} = \frac{P}{\pi ab}$$

$$\frac{2\pi}{P}(t - t_0) = \mathcal{E} - e \sin(\mathcal{E})$$

Apply Kepler's 3rd Law $P = 2\pi \sqrt{\frac{a^3}{\mu}}$

$$\underbrace{\sqrt{\frac{\mu}{a^3}}}_{\text{mean anomaly}} (t - t_0) = \mathcal{E} - e \sin(\mathcal{E})$$

mean anomaly

$$\boxed{M = \mathcal{E} - e \sin(\mathcal{E})} \quad \text{Kepler's eq.}$$

We recall the constant $\sqrt{\frac{\mu}{a^3}} = \text{mean motion}$

$$\boxed{n = \sqrt{\frac{\mu}{a^3}} \quad M = n(t - t_0)}$$

For this equation to be of much use, we need to relate true anomaly, θ , to eccentric anomaly, $\mathcal{E}$.

$$\sin \theta = \frac{\sqrt{1-e^2} \sin(\mathcal{E})}{1 - e \cos(\mathcal{E})}$$

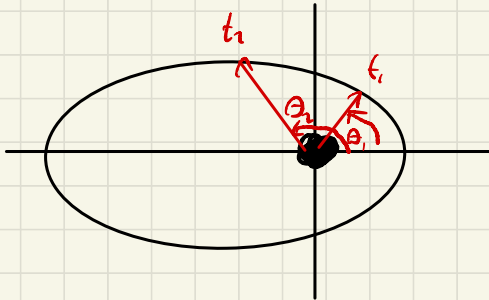
$$\cos(\theta) = \frac{\cos(\mathcal{E}) - e}{1 - e \cos(\mathcal{E})}$$

$$\sin(\mathcal{E}) = \frac{\sqrt{1-e^2} \sin(\theta)}{1 + e \cos(\theta)}$$

$$\cos(\mathcal{E}) = \frac{\cos(\theta) + e}{1 + e \cos \theta}$$

$$\tan\left(\frac{\xi}{2}\right) = \sqrt{\frac{1-e}{1+e}} \tan\left(\frac{\theta}{2}\right)$$

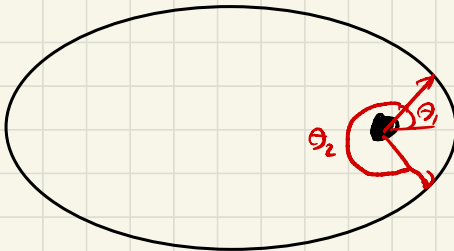
The direct application of Kepler's equation allows us to find the time required to travel between two points on orbit.



$$M = \xi - e \sin(\xi)$$

$$M = n(t - t_0) = \xi - e \sin(\xi)$$

$$n(t_2 - t_1) = (\xi_2 - e \sin(\xi_2)) - (\xi_1 - e \sin(\xi_1))$$



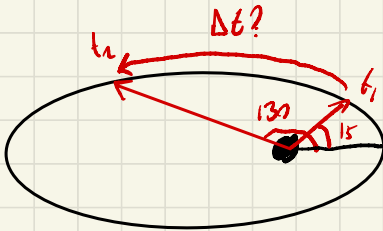
if we go through periaapsis k times...

$$n(t_2 - t_1) = 2\pi k + (\xi_2 - e \sin(\xi_2)) - (\xi_1 - e \sin(\xi_1))$$

$$n = \sqrt{\frac{\mu}{a^3}}$$

$$(t_2 - t_1) = \sqrt{\frac{a^3}{\mu}} \left(2\pi k + (\xi_2 - e \sin \xi_2) - (\xi_1 - e \sin \xi_1) \right)$$

Example: Consider a spacecraft of altitude of 410 km at r_p and $e = 0.3$, Now long from 15° to 130° ?



$$\cos(\xi_1) = \frac{e + \cos \theta}{1 + e \cos \theta}$$



$$\xi_1 = 0.1926 \text{ rad}$$

$$\xi_2 = 2.0094 \text{ rad}$$

$$r_p = h + r_e = 6378 \text{ km} + 410 \text{ km} = 6788 \text{ km}$$

$$r_p = a(1 - e) \rightarrow a = 9697.1 \text{ km}$$

$$t_2 - t_1 = \sqrt{\frac{a^3}{\mu}} \left((\xi_2 - e \sin(\xi_2)) - (\xi_1 - e \sin(\xi_1)) \right)$$

$$\Delta t = 2424 \text{ sec or } 40.4 \text{ min}$$

Kepler's problem

Kepler's problem is the task of finding the location of a body in orbit after time has passed

$$(\text{solve Kepler's eq}) \quad M = n(t - t_0) = \xi - e \sin(\xi)$$

This is a transcendental function (Non-algebraic)

So, in general we must use a numerical method.

The most common solution uses a Newton-Rapson iteration scheme.

Therefore, let's rewrite Kepler's eq. -

$$F(\xi) = \xi - e \sin(\xi) - M = 0$$

Assume we have a good initial guess of ξ

$$\xi = \xi_n + \delta_n, \text{ where } \delta_n \text{ is small}$$

We can take a Taylor series expansion...

$$F(\xi) = F(\xi_n + \delta_n) = F(\xi_n) + F'(\xi_n)\delta_n + \underbrace{\frac{1}{2}F''(\xi_n)\delta_n^2}_{\text{ignore}} \dots$$

The only first order term is δ_n

$$0 = F(\xi) \approx F(\xi_n) + f'(\xi_n)\delta_n$$

$$f'(\xi_n)\delta_n = -f(\xi_n)$$

$$\delta_n = -\frac{f(\xi_n)}{f'(\xi_n)}$$

$$\begin{aligned} f(\xi_n) &= \xi_n - e \sin(\xi_n) - M \\ f'(\xi_n) &= 1 - e \cos(\xi_n) \end{aligned}$$

$$S_n = - \frac{\xi_n - e \sin(\xi_n) - M}{1 - e \cos(\xi_n)} = \frac{M - \xi_n + e \sin(\xi_n)}{1 - e \cos(\xi_n)}$$

$$\text{Thus, } \xi_{n+1} = \xi_n + S_n$$

The trick is to find a good initial guess for ξ_0

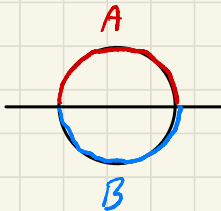
For a circle orbit ($e=0$) $\rightarrow M = \xi$

Assume we've already reduced mean anomaly to

$$-2\pi < M < 2\pi$$

A good initial guess of ξ_0 is

$$\xi = \begin{cases} M - e, & -\pi < M < 0 \text{ or } M > \pi \\ M + e, & \text{otherwise} \end{cases}$$



If the orbit is not an ellipse, we must use alternate formulations of Kepler's eqn.

$$t - t_0 = \frac{1}{2\sqrt{\mu}} \left[\rho D + \frac{1}{3} D^3 \right] \quad D = \sqrt{\rho} \tan\left(\frac{\theta}{2}\right)$$

↑
Bartels eqn. only for $e=1$

Hyperbolic.

$$t - t_0 = \sqrt{\frac{-a}{\mu}}^3 (e \sinh(F) - F) \quad \cosh(F) = \frac{e * \cos\theta}{1 + e \cos\theta}$$

Be careful, if numerically $e \rightarrow 1$

All of this may be avoided using Universal variables...

↳ we get a lot less intuitive insight.

Universal variables

Recall $\epsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$

Some algebra will yield

$$\dot{r}^2 = -\frac{\mu p}{r^2} + \frac{2\mu}{r} - \frac{\mu}{a}$$

We'll now introduce a change of variables as a Sundman transformation

$$\frac{dx}{dt} = \frac{\sqrt{h}}{r} \rightarrow x \text{ is a new variable.}$$

$$\dot{r}^2 = \left(\frac{dr}{dx} \frac{dx}{dt} \right)^2 = \left(\frac{dr}{dx} \right)^2 \frac{h}{r^2} = -\frac{h}{r^2} + \frac{2h}{r} - \frac{h}{a}$$

$$\left(\frac{dr}{dx} \right)^2 = -p + 2r - \frac{r^2}{a}$$

$$dx = \frac{dr}{\sqrt{-p + 2r - \frac{r^2}{a}}}$$

↓ integrate

$$x + C_0 = \sqrt{a} \sin^{-1} \left(\frac{\frac{r}{a} - 1}{\sqrt{1 - p/a}} \right) \rightarrow p = a(1 - e^2) \rightarrow e = \sqrt{1 - p/a}$$

$$r = a \left[1 + e \sin \left(\frac{x + C_0}{\sqrt{a}} \right) \right]$$

So, what is the meaning of x ?

$$r = a(1 - e \cos(\xi)) \Rightarrow \sin \left(\frac{x + C_0}{\sqrt{a}} \right) = -\cos(\xi)$$

Aside: $\sin(\alpha) = \cos\left(\frac{\pi}{2} \pm \alpha\right)$

$$\sin\left(\frac{x+c_0}{\sqrt{a}}\right) = -\cos\left(\frac{\pi}{2} + \frac{x+c_0}{\sqrt{a}}\right) = -\cos(\xi)$$

$$\text{Thus, } \xi = \frac{\pi}{2} + \frac{x+c_0}{\sqrt{a}} \quad \text{Let } x=0 \text{ at } t_1 \rightarrow \xi_1 = \frac{\pi}{2} + \frac{c_0}{\sqrt{a}}$$

$$\text{Then, } \xi = \frac{x}{\sqrt{a}} + \xi_1$$

$$\text{Thus, } x = \sqrt{a} (\xi - \xi_1) \rightarrow x = \sqrt{a} (\Delta\xi)$$

Backs to the main story...

Substitute into the def of $\dot{x}$

$$\frac{dx}{dt} = \frac{\sqrt{\mu}}{r} \Rightarrow \sqrt{\mu} dt = r dx$$

$$\sqrt{\mu} dt = a \left[1 + e \sin\left(\frac{x+c_0}{\sqrt{a}}\right) \right] dx$$

Integrate... $x=0$ at t_1

$$\Delta t \sqrt{\mu} = ax - ae\sqrt{a} \left(\cos\left(\frac{x+c_0}{\sqrt{a}}\right) - \cos\left(\frac{c_0}{\sqrt{a}}\right) \right)$$

$$\sqrt{\mu} \Delta t = a \left(x - \sqrt{a} \sin\left(\frac{x}{\sqrt{a}}\right) + \frac{\hat{r}_1 \cdot \vec{v}_1}{\sqrt{\mu}} a \left(1 - \cos\left(\frac{x}{\sqrt{a}}\right) \right) + r_1 \sqrt{a} \sin\left(\frac{x}{\sqrt{a}}\right) \right)$$

Apply to $r \dots$

$$r = a + a \left[\frac{\vec{r}_i \cdot \vec{v}_i}{\sqrt{\mu a}} \sin\left(\frac{x}{\sqrt{a}}\right) + \left(\frac{r_i}{a} - 1\right) \cos\left(\frac{x}{\sqrt{a}}\right) \right]$$

Still have some problems when $a \rightarrow \infty$ or $a < 0$

Define another new variable.

$$z = \frac{x^2}{a} \rightarrow a = \frac{x^2}{z}$$

$$\text{Aside: } z = \frac{1}{a} (\sqrt{a} \Delta \epsilon)^2 = (\Delta \epsilon)^2$$

Such that

$$\sqrt{\mu} \Delta t = \left[\frac{\sqrt{z} - \sin \sqrt{z}}{\sqrt{z}^3} \right] x^3 + \frac{\vec{r}_i \cdot \vec{v}_i}{\sqrt{\mu}} x^2 \left[\frac{1 - \cos(\sqrt{z})}{z} \right] + \frac{r_0 x \sin(\sqrt{z})}{\sqrt{z}}$$

$$r = \frac{x^2}{z} + \frac{\vec{r}_i \cdot \vec{v}_i}{\sqrt{\mu}} \cdot \frac{x}{\sqrt{z}} \cdot \sin(\sqrt{z}) + r_0 \cos(\sqrt{z}) - \frac{x^2}{z} \cos \sqrt{z}$$

observe...

$$\begin{aligned} C(z) &= \frac{1 - \cos(\sqrt{z})}{z} = \frac{1 - \cosh(\sqrt{-z})}{z} = \frac{1}{z!} - \frac{z}{4!} + \frac{z^3}{6!} - \frac{z^5}{8!} \\ &= \sum_{k=0}^{\infty} \frac{(-z)^k}{(2k+2)!} \end{aligned}$$

$$s(z) = \frac{\sqrt{z} - \sin(\sqrt{z})}{\sqrt{z}^3} = \frac{\sinh(\sqrt{-z}) - \sqrt{-z}}{\sqrt{(-z)^3}} = \sum_{k=0}^{\infty} \frac{(-z)^k}{(2k+3)!}$$

and Finally

$$\Gamma_{\mu} \Delta t = x^3 S + \frac{\bar{r} \cdot \bar{v}}{\Gamma_{\mu}} x^2 C + r_1 x (1 - 2S)$$

$$r = \Gamma_{\mu} \frac{dt}{dx} = x^2 C + \frac{\bar{r} \cdot \bar{v}}{\Gamma_{\mu}} x (1 - 2S) + r_1 (1 - 2C)$$

at $\bar{r}_1, \bar{v}_1 \rightarrow x=0$ at t_1 ,

thus, the Kepler problem has $\Delta t = t_2 - t_1$ and we know t_2

Use Newton-Raphson to update x

$$x_{n+1} = x_n + \frac{\Delta t - \Delta t_n}{\left. \frac{dt}{dx} \right|_{x=x_n}}$$

The only missing part is a good initial guess for x

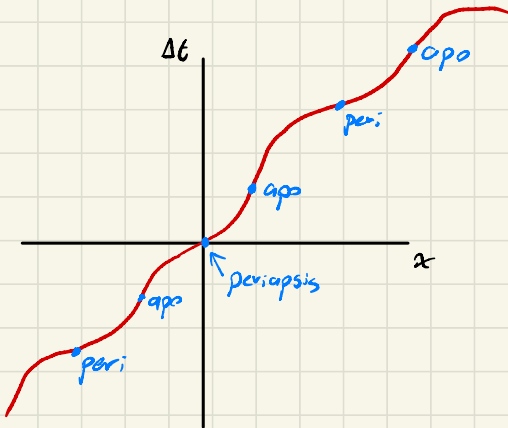
for elliptical orbits: $\frac{\Gamma_{\mu}}{a} (t_2 - t_1) = x$

in BMWS eq. 4-76 for hyperbolas

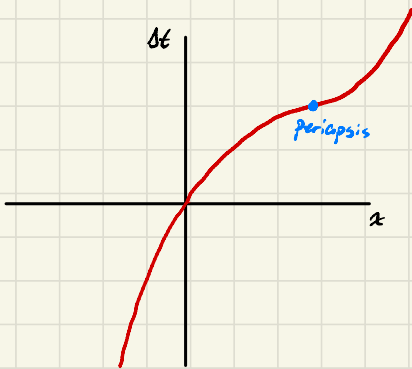
This all quickly converges, t vs. x curve is well-behaved.

Recall $\frac{dt}{dx} = \frac{r}{\Gamma_{\mu}} \leftarrow$ maximum at apapsis
minimum at periapsis

Elliptical orbit at periastris at t_1



Hyperbolic with $\bar{r}_0$ before periastris



End of Material for Test 1

The Lagrange Coefficients (f and g functions)

Orbits are planar, and spanned by $\bar{r}$ and $\bar{v}$ at any given time

Thus, the position and velocity at any later time may be written as a linear combination of position and velocity at a reference time.

$$\bar{\mathbf{r}} = f\bar{\mathbf{r}}_0 + g\bar{\mathbf{v}}_0$$

$$\bar{\mathbf{v}} = \dot{f}\bar{\mathbf{r}}_0 + \dot{g}\bar{\mathbf{v}}_0$$

We can write position and velocity in the perifocal frame

$$\bar{\mathbf{r}} = x\hat{\mathbf{p}} + y\hat{\mathbf{q}}$$

$$\bar{\mathbf{v}} = \dot{\bar{\mathbf{r}}} = \dot{x}\hat{\mathbf{p}} + \dot{y}\hat{\mathbf{q}}$$

$$\hat{\mathbf{h}} = \bar{\mathbf{r}} \times \bar{\mathbf{v}} = (x\hat{\mathbf{p}} + y\hat{\mathbf{q}}) \times (\dot{x}\hat{\mathbf{p}} + \dot{y}\hat{\mathbf{q}}) = (x\dot{y} - \dot{x}y)\hat{\omega}$$

Let us now solve for $\hat{\mathbf{p}}$ and $\hat{\mathbf{q}}$ in the terms of $\bar{\mathbf{r}}$ and $\bar{\mathbf{v}}$ at the same reference time t_0

$$\bar{\mathbf{r}}_0 = x_0\hat{\mathbf{p}} + y_0\hat{\mathbf{q}} \rightarrow \hat{\mathbf{q}} = \frac{1}{y_0}\bar{\mathbf{r}}_0 - \frac{x_0}{y_0}\hat{\mathbf{p}}$$

$$\bar{\mathbf{v}}_0 = \dot{x}_0\hat{\mathbf{p}} + \dot{y}_0\hat{\mathbf{q}} = \dot{x}_0\hat{\mathbf{p}} + \dot{y}_0\left(\frac{1}{y_0}\bar{\mathbf{r}}_0 - \frac{x_0}{y_0}\hat{\mathbf{p}}\right) = \underbrace{\left(\frac{x_0\dot{x}_0 - \dot{x}_0x_0}{y_0}\right)}_{-|\hat{\mathbf{h}}|}\hat{\mathbf{p}} + \frac{\dot{y}_0}{y_0}\bar{\mathbf{r}}_0$$

$$\bar{\mathbf{v}}_0 = -\frac{h}{y_0}\hat{\mathbf{p}} + \frac{\dot{y}_0}{y_0}\bar{\mathbf{r}}_0$$

$$\rightarrow \frac{h}{y_0}\hat{\mathbf{p}} = \frac{\dot{y}_0}{y_0}\bar{\mathbf{r}}_0 - \bar{\mathbf{v}}_0 \Rightarrow \hat{\mathbf{p}} = \frac{\dot{y}_0}{h}\bar{\mathbf{r}}_0 - \frac{y_0}{h}\bar{\mathbf{v}}_0$$

and

$$\hat{\mathbf{q}} = \frac{1}{y_0}\bar{\mathbf{r}}_0 - \frac{x_0}{y_0}\hat{\mathbf{p}} = \hat{\mathbf{q}} = \frac{1}{y_0}\bar{\mathbf{r}}_0 - \frac{x_0}{y_0}\left(\frac{\dot{y}_0}{h}\bar{\mathbf{r}}_0 - \frac{y_0}{h}\bar{\mathbf{v}}_0\right)$$

Simplify to...

$$\hat{q} = \frac{h - x_0 \dot{y}_0}{y_0 h} \bar{r}_0 + \frac{x_0}{h} \bar{v}_0$$

$$\hat{q} = \frac{-\cancel{y_0} \dot{x}_0}{\cancel{y_0} h} \bar{r}_0 + \frac{x_0}{h} \bar{v}_0 \rightarrow \hat{q} = -\frac{\dot{x}_0}{h} \bar{r}_0 + \frac{x_0}{h} \bar{v}_0$$

$$\bar{r} = x \left(\frac{y_0}{h} \bar{r}_0 - \frac{x_0}{h} \bar{v}_0 \right) - y \left(-\frac{\dot{x}_0}{h} \bar{r}_0 + \frac{x_0}{h} \bar{v}_0 \right)$$

$$\bar{r} = \left(\frac{x y_0 - y \dot{x}_0}{h} \right) \bar{r}_0 + \left(\frac{-x x_0 + y \dot{x}_0}{h} \right) \bar{v}_0$$

$$\bar{v} = \left(\frac{\dot{x} y_0 - y \dot{x}_0}{h} \right) \bar{r}_0 + \left(\frac{\dot{x} y_0 + y x_0}{h} \right) \bar{v}_0$$

so, $\bar{r} = f \bar{r}_0 + g \bar{v}_0$	where $f = \frac{x y_0 - y \dot{x}_0}{h}$	$g = \frac{-x x_0 + y \dot{x}_0}{h}$
$\bar{v} = \dot{f} \bar{r}_0 + \dot{g} \bar{v}_0$	$\dot{f} = \frac{\dot{x} y_0 - y \dot{x}_0}{h}$	$\dot{g} = \frac{-\dot{x} y_0 + y x_0}{h}$

recall that $\bar{h} = \bar{r} \times \bar{v}$

$$\bar{h} = (f\bar{r}_0 + g\bar{v}_0) \times (f\bar{r}_0 + g\bar{v}_0) = fg(\bar{r}_0 \times \bar{v}_0) + gf(\bar{v}_0 \times \bar{r}_0)$$

$$\bar{h} = (fg - gf) \underbrace{(\bar{r}_0 \times \bar{v}_0)}_{\bar{h}} \quad \bar{h} = \underbrace{(fg - gf)}_1 \bar{h}$$

Thus, $\boxed{fg - gf = 1}$, thus $f, g, \dot{f}, \dot{g}$ are not independent

We can write $f, g, \dot{f}, \dot{g}$ in terms of $\Delta\theta, \Delta\epsilon, x$

$$f = 1 - \frac{a}{r_0} (1 - \cos(\Delta\epsilon)) \quad \dot{f} = -\sqrt{\mu} a \sin(\Delta\epsilon)$$

$$g = \Delta t - \sqrt{\frac{a^3}{\mu}} (\Delta\epsilon - \sin(\Delta\epsilon)) \quad \dot{g} = 1 - \frac{a}{r} (1 - \cos(\Delta\epsilon))$$

$$f = 1 - \frac{x^2}{r_0} C$$

$$\dot{f} = \frac{\sqrt{\mu}}{r r_0} x (2s - 1)$$

$$g = \Delta t - \frac{x^3}{\sqrt{\mu}} S$$

$$\dot{g} = 1 - \frac{x^2}{r} C$$

These let you propagate $\bar{r}_0$ and $\bar{v}_0$ directly.

Lambert's problem (Gauss's problem)

Given two position vectors ($\vec{r}_1$ and $\vec{r}_2$) and the elapsed time, Δt , find the orbit that matches the boundary conditions.

If $\vec{r}_1 = \vec{r}(t_1)$ and $\vec{r}_2 = \vec{r}(t_1 + \Delta t)$ are not colinear, they define a plane.

Let $\Delta\theta = \theta_2 - \theta_1$ be the difference in true anomaly between $\vec{r}_1$ and $\vec{r}_2$.

$$\cos(\Delta\theta) = \frac{(\vec{r}_1 \cdot \vec{r}_2)}{|\vec{r}_1| |\vec{r}_2|}$$

↑
quadrant ambiguous

for any pair $\vec{r}_1$ and $\vec{r}_2$, two possible solutions

1) short way $0 < \Delta\theta < \pi$

2) long way $\pi < \Delta\theta < 2\pi$

Also need specified direction of motion.